

Cost-Asymmetric Physics-Informed Forecasting for Robust NHS Critical-Care Surge Allocation

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Abstract. NHS critical-care planning during epidemic waves must pre-position surge capacity and route patients across regions before demand is known. We propose a forecast-to-decision pipeline that couples a per-region physics-informed forecaster on the $SEI_a I_s HCRD$ system to a tail-risk robust linear program for bed-surge allocation. The forecaster is trained with a cost-asymmetric multi-quantile pinball loss whose $q^{(0.9)}$ branch penalises under-prediction nine-to-one, together with a learnable level-and-trend anchor and a leakage-free per-origin recalibration that removes a long-horizon over-prediction bias. On a seven-region NHS England case study (Aug 2020 to Aug 2022) it cuts 14-day RMSE by 25% over per-origin refitted ARIMA and by 42% over a neural GRU, and is competitive across all horizons (best at 7–21 days and level with refitted ARIMA at 28, where seasonal-naive is strongest). Its quantile forecasts drive a hybrid-regional allocation solved exactly as an LP. With unconstrained transfers all budget-exhausting policies tie on coverage and the tail-risk robust formulation wins on transfer burden; once transfers are constrained the tie breaks and the proportional heuristics concede up to 60% more expected unmet demand. A sensitivity analysis over the forecaster, budget, tail-risk weight, travel cap, and transfer cap shows the budget to be the dominant control.

Keywords: physics-informed neural networks, cost-asymmetric forecasting, pinball loss, robust optimisation, linear programming, NHS England

1 Introduction

NHS critical-care surge planning during epidemic waves balances the cost of pre-positioning capacity against the risk of unmet demand while keeping inter-regional transfers manageable across the seven NHS England regions. COVID-19 placed sustained pressure on adult mechanical-ventilation capacity, making this a coupled forecasting and allocation problem under regime shift. Existing forecasting work spans statistical and recurrent baselines [2], graph-based epidemic and hospital-demand models [9,8], and physics-informed neural networks [15,1,14]; interval and quantile evaluation is also standard in epidemic forecast hubs [4]. The closest ICU-occupancy comparator, a bidirectional LSTM with lagged mobility features and MC-dropout intervals on Lazio data [17], like most such work targets a single health system rather than multi-region planning. These forecasters are usually

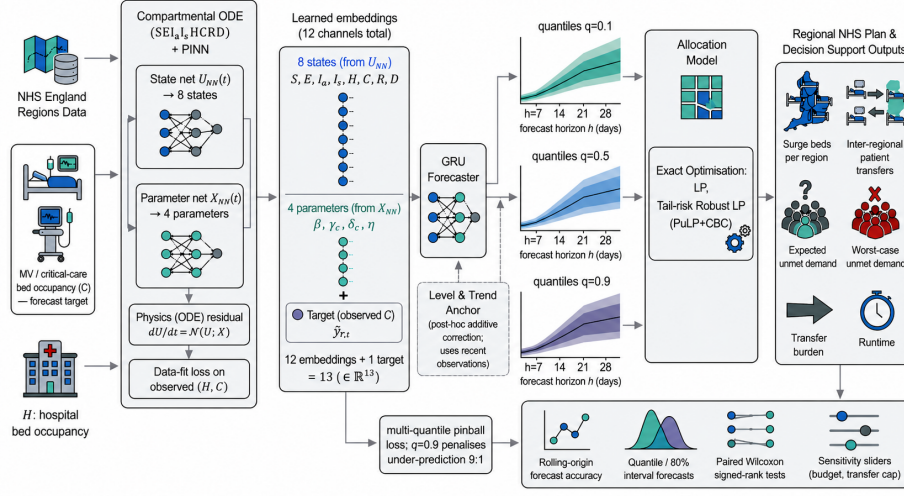


Fig. 1. Pipeline for NHS critical-care surge planning, from regional data and physics-informed forecasting to scenario-based robust allocation and evaluation.

optimised for symmetric statistical accuracy, although critical-care planning has an asymmetric error cost: under-prediction risks stranded patients, whereas over-prediction wastes prepared capacity.

On the allocation side, robust facility-location and health-resource models handle demand uncertainty with MILP, two-stage optimisation, and metaheuristics [7,18,16,5]. Bertsimas et al. [3] couple DELPHI dynamics to a bilinear MIP for vaccination-site location, while Luo and Stellato [11] embed neural-ODE compartmental dynamics inside a McCormick-relaxed facility-location model. Our setting differs from Smart Predict-then-Optimize [6]. The forecaster is not trained through allocation regret, but its multi-quantile pinball objective is cost-asymmetric and feeds a robust optimisation layer. To our knowledge, no prior work combines physics-informed forecasting of NHS regional critical-care demand with cost-asymmetric quantile training and an exact robust linear-programming allocation across the seven NHS England regions.

This paper presents a forecast-to-decision pipeline (Figure 1) in four phases. *Phase A* pre-trains a per-region PINN on the $SEI_a I_s HCRD$ ODE system; its twelve outputs (eight compartmental states, four transmission and clinical parameters) are frozen and passed to a temporal GRU head that emits the horizon-wise quantiles $q^{(0.1)}$, $q^{(0.5)}$, $q^{(0.9)}$ under a multi-quantile pinball loss whose $q^{(0.9)}$ branch penalises under-prediction nine-to-one, with a level-and-trend anchor. *Phase B* discretises these quantiles into three scenarios with weights $\pi = (0.2, 0.6, 0.2)$. *Phase C* solves the allocation exactly as a linear program (deterministic and tail-risk-robust) via PuLP+CBC against three heuristic baselines on a shared LP-slave oracle. *Phase D* reports operational metrics and a sensitivity analysis over the forecaster, budget B , tail-risk weight λ_3 , travel cap τ , and transfer cap κ .

The main contributions are as follows: (1) **Cost-asymmetric forecaster.** A multi-quantile pinball loss, a level-and-trend anchor, and a leakage-free per-origin recalibration reduce $h=14$ RMSE by 25% relative to refitted ARIMA and 42% relative to a per-region GRU, and make the model competitive across all horizons (best at $h=7-21$ and level with refitted ARIMA at $h=28$, where seasonal-naive leads). (2) **Robust LP allocation.** With unconstrained transfers the hybrid-regional robust LP ties every budget-exhausting policy on coverage and wins on transfer burden ($4.6\times$ below population-proportional); constraining transfers (sparser network or capped transfer service) breaks the tie, the proportional rules concede up to 60% more expected unmet demand, and the robust LP is never beaten on coverage.

2 Forecasting Module

2.1 SEIRD Compartmental Model

For each region $r \in \mathcal{R}$ with population N_r , we model the local epidemic with an eight-compartment $SEI_a I_s HCRD$ system whose states are susceptible, exposed, asymptomatic/symptomatic infectious, hospitalised, critical-care, recovered, and deceased. Figure 2 shows the flow topology, in which the distinct probabilities ω ($I_s \rightarrow H$) and ϕ ($H \rightarrow C$) disambiguate the two clinical thresholds. The system evolves as

$$\begin{aligned} \dot{S}_r &= -\beta_r S_r (I_{s,r} + I_{a,r}) / N_r + \eta_r R_r, & \dot{E}_r &= \beta_r S_r (I_{s,r} + I_{a,r}) / N_r - \alpha E_r, \\ \dot{I}_{s,r} &= \alpha \rho E_r - d_s I_{s,r}, & \dot{I}_{a,r} &= \alpha (1 - \rho) E_r - d_a I_{a,r}, \\ \dot{H}_r &= d_s \omega I_{s,r} - (d_H + \mu) H_r, & \dot{C}_r &= \phi d_H H_r - (\gamma_{c,r} + \delta_{c,r}) C_r, \\ \dot{R}_r &= d_s (1 - \omega) I_{s,r} + d_a I_{a,r} + (1 - \phi) d_H H_r + \gamma_{c,r} C_r - \eta_r R_r, \\ \dot{D}_r &= \mu H_r + \delta_{c,r} C_r, \end{aligned} \tag{1}$$

subject to $S_r + E_r + I_{a,r} + I_{s,r} + H_r + C_r + R_r + D_r = N_r$. Here $\alpha, d_s, d_a, d_H, \rho, \omega, \mu, \phi$ are fixed clinical constants (values from [1], Table 1) and $\beta_r, \gamma_{c,r}, \delta_{c,r}, \eta_r$ are time-varying region-specific parameters (transmission, critical-care recovery and death, and immunity waning) learned from data.

2.2 Per-Region PINN-SEIRD

The PINN consists of two MLPs following [1]: a state network U_{NN}^r (five hidden layers of 20 units) and a parameter network X_{NN}^r for $(\beta_r, \gamma_{c,r}, \delta_{c,r}, \eta_r)$ (three hidden layers of 20 units), with tanh hidden activations and sigmoid outputs. The loss combines a data fit on the two observable compartments with an ODE residual against Equation (1). The observed compartments are H (daily general-and-acute hospital bed occupancy) and C (mechanical-ventilation, i.e. critical-care, bed occupancy), the only states with daily regional NHS counts; C is the quantity

the temporal head forecasts and the allocation consumes as demand ξ :

$$\begin{aligned} \mathcal{L}_r^{PINN} = & \sum_{t \in \mathcal{T}_{\text{train}}} \sum_{k \in \{H, C\}} (U_{NN,k}^r(t) - y_{r,t,k}/N_r)^2 \\ & + \lambda_{ode} \sum_{t \in \mathcal{T}_{\text{coll}}} \left\| \frac{dU_{NN}^r(t)}{dt} - \mathcal{N}(U_{NN}^r; X_{NN}^r) \right\|^2, \end{aligned} \quad (2)$$

where $\mathcal{N}$ is the right-hand side of Equation (1), derivatives use PyTorch autograd at 256 collocation points sampled uniformly in $\mathcal{T}_{\text{coll}}$, and observations are population-normalised. We pre-train each region’s PINN for 1500 Adam steps with $\lambda_{ode} = 0.1$, $\text{lr} = 5 \times 10^{-3}$; both terms converge to the $2\text{--}8 \times 10^{-6}$ range across regions.

With only H and C observed regionally, the four learnable parameters $(\beta_r, \gamma_{c,r}, \delta_{c,r}, \eta_r)$ are not jointly identifiable ($\gamma_{c,r}$ and $\delta_{c,r}$ appear additively in $\dot{C}_r$), so we make no interpretability claim for individual trajectories: the four-channel output is consumed as a learned feature vector by the GRU, whose signal the “w/o PINN params” ablation confirms (Section 5.1; $h=14$ RMSE $14.46 \rightarrow 27.25$). Structural identifiability under $\{H, C\}$ observation is future work.

2.3 Temporal Head with Level-and-Trend Anchor

Once pre-trained the PINN is *frozen*: at every origin, including those in the Omicron test period, its networks are *evaluated* (never refit) to supply features, so any temporal extrapolation of the state network beyond its Alpha/Delta fit is absorbed and corrected downstream by the GRU and the anchor of Equation (3). For each lookback window of length $L = 28$ ending at origin t , the augmented feature sequence is $\mathbf{f}_{r,i} = [U_{NN}^r(i) \parallel X_{NN}^r(i) \parallel \tilde{y}_{r,i}] \in \mathbb{R}^{13}$, $i \in \{t-L+1, \dots, t\}$. Here $U_{NN}^r(i)$ contributes all eight compartments (including the latent S, E, I_a, I_s, R, D that are never directly observed), so the head receives the full physics-inferred state rather than the two fitted channels alone, and $\tilde{y}_{r,i}$ is the observed target z-scored with training-period statistics only (no test-set leakage). A two-layer GRU (hidden 128, dropout 0.15) decodes the final hidden state via a per-horizon

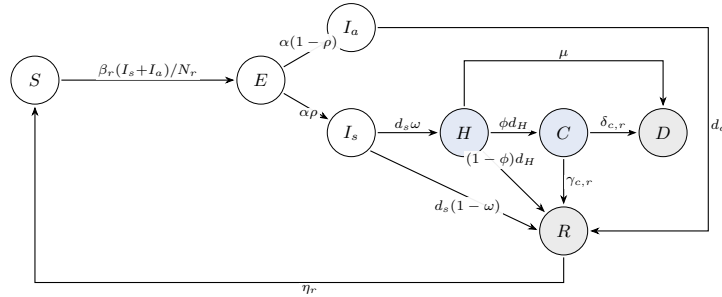


Fig. 2. $SEI_a I_s HCRD$ compartmental model used by the physics-informed forecaster.

two-layer MLP (width 64, GELU) into standardised quantile deltas $\Delta_{r,t}^{h,q}$ for $h \in \mathcal{H} = \{7, 14, 21, 28\}$ and $q \in \mathcal{Q} = \{0.1, 0.5, 0.9\}$.

A *level-and-trend anchor* makes the GRU’s job residual:

$$\hat{\zeta}_{r,t}^{h,q} = \Delta_{r,t}^{h,q} + a^{h,q} \tilde{y}_{r,t} + g^{h,q} v_{r,t}^{(7)} \left(\frac{h}{7}\right)^{\psi^q}, \quad (3)$$

where $\tilde{y}_{r,t}$ is the most recent z-scored observation, $v_{r,t}^{(7)} = \tilde{y}_{r,t} - \tilde{y}_{r,t-7}$ is the one-week slope, and $a^{h,q}, g^{h,q}, \psi^q \in (0, 1)$ are learnable scalars. The Holt-Winters-style damping exponent ψ^q prevents slope-extrapolation overshoot at long horizons; we initialise $\psi^{0.5} \approx 0.88$ (near-linear) and $\psi^{0.1} = \psi^{0.9} = 0.5$ (square-root damping). Forecasts in the original scale follow by inverting standardisation.

2.4 Pinball Loss for Cost-Asymmetric Quantiles

The forecaster minimises the multi-quantile pinball loss [10]:

$$\mathcal{L}_{\text{pinball}} = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \mathbb{E}_{(t,h)} [\max(qe, (q-1)e)], \quad e = \tilde{y}_{r,t+h-1} - \hat{\zeta}_{r,t}^{h,q}, \quad (4)$$

evaluated at all $(h, q) \in \mathcal{H} \times \mathcal{Q}$. The $q = 0.5$ pinball collapses to L1 (point accuracy); the $q = 0.9$ pinball penalises under-prediction nine-to-one. Splitting the kernel into its two arms, $\max(qe, (q-1)e) = q \max(e, 0) + (1-q) \max(-e, 0)$, exposes it as a convex newsvendor surrogate of the allocation’s positive-part shortage cost, with the $q=0.9$ asymmetry encoding that an under-bed risks a stranded patient whereas an over-bed only wastes pre-positioned capacity [6]. The forecaster is not trained through the allocation solver. We report $\hat{y}^{0.5}$ as the point forecast and $[\hat{y}^{0.1}, \hat{y}^{0.9}]$ as an 80% interval.

We train for up to 800 AdamW epochs (lr = 10^{-3} , weight decay 10^{-4} , cosine annealing) on one-day-stride Alpha-wave lookback windows (270 per region, 1,890 in total), with early stopping (patience 80) on the 1,134 Delta-wave validation windows’ pinball loss. Once selected, the model is frozen and applied to the Omicron test period via $\Delta t = 7$ -day rolling origins, yielding 32 origin dates.

2.5 Per-Origin Recalibration

Applied frozen across a regime shift (high-demand Alpha/Delta training, mild Omicron test), the point forecast inherits a systematic over-prediction that grows with horizon. We correct it with two leakage-free steps. The first recalibrates the bias, subtracting at each rolling origin t and horizon h the mean of the model’s last $K=3$ h -step errors whose target date is already observed ($< t$), the adaptation that a per-origin refit gives for free. The second blends with persistence, mixing the recalibrated point forecast with seasonal-naive(7) under a per-horizon weight $\alpha \in \{0, 0.1, \dots, 1\}$ chosen on an expanding window of realised past origins (fitted once at least seven origin–region pairs are realised; until then $\alpha=1$, pure model), so long horizons lean on persistence. Both use only information available at t , and the 80% interval is left as produced by the quantile heads, so the downstream allocation is unchanged.

3 Regional Allocation Model

The allocation module decides how many surge beds to place in each NHS region and how to route excess demand through inter-regional patient transfers, given the quantile forecasts of Section 2. We present a deterministic linear program, a tail-risk robust extension, and closed-form heuristic baselines.

3.1 Sets, Parameters, and Decision Variables

The scope is *hybrid regional*: demand nodes and capacity sites coincide with the seven NHS England commissioning regions, one set $\mathcal{R} = \{r_1, \dots, r_7\}$. Unlike facility-location models that choose which sites to open [3], every region already operates critical care, so there is no opening decision and per-bed surge cost is uniform; the model decides only how many beds per region and how to route patients. Indices: regions $r, r' \in \mathcal{R}$; horizons $h \in \mathcal{H} = \{7, 14, 21, 28\}$ days; scenarios $s \in \mathcal{S} = \{\text{low}, \text{median}, \text{high}\}$ from $q^{(0.1)}, q^{(0.5)}, q^{(0.9)}$ with weights $\pi = (0.2, 0.6, 0.2)$, a tractable representative-quantile discretisation of lower, central, and upper demand stress points; the weights define the optimisation scenario mixture, not calibrated probability bins. Parameters: forecast demand $\xi_{r,h}^s = \hat{q}_r^{(s)}(t_0 + h)$; baseline critical-care capacity $\Gamma_r = 1.05 \times \text{Delta-peak}$ (the bed ceiling of region r); per-region surge cap $K_r = 0.5 \Gamma_r$; budget $B = 0.20 \sum_r \Gamma_r$; great-circle inter-centroid distance $\ell_{rr'}$ in km ($\times 1.3$ road-detour); travel time $T_{rr'} = 60 \ell_{rr'} / 80$ min; mutual-aid cap $\tau = 240$ min; outbound transfer cap κ (displaced patients a region's transfer service can sustain at a checkpoint; uncapped at the headline operating point). Decision variables fall in two layers. The *first-stage* (here-and-now) capacity decision is the surge allocation $b_r \in [0, K_r]$, shared across scenarios and horizons (beds stood up once serve every weekly checkpoint). This is the planner's only true control. The per-scenario *second-stage* recourse is the flow $z_{rr',h}^s \geq 0$ (defined for $T_{rr'} \leq \tau$), whose self-loop $z_{rr,h}^s$ is demand served *locally* at zero distance and whose off-diagonal ($r' \neq r$) entries are inter-regional transfers, together with the unmet demand $u_{r,h}^s \geq 0$; a tail auxiliary $W \geq 0$ completes the robust variant. Of the 42 ordered region pairs, 24 satisfy $T_{rr'} \leq \tau$ (12 undirected mutual-aid links); Midlands reaches every region, South West only Midlands.

3.2 Deterministic Median-Scenario LP

The deterministic model optimises the median scenario m ($q^{(0.5)}$):

$$\min \sum_{r,h} u_{r,h}^m + \theta \sum_{r \neq r',h} \ell_{rr'} z_{rr',h}^m, \quad (5)$$

with $\theta = 10^{-3}$ a transfer tie-break weight (one bed of weekly snapshot unmet outweighs 1000 patient-km of transfer). The objective minimises *unmet demand*; the second term only breaks ties between equal-coverage plans by preferring less patient transport, so the surge beds b are placed to reduce shortage, not to minimise transfers. Each $u_{r,h}^m$ is a snapshot of bed-shortfall at the weekly

checkpoint h , so the objective is the snapshot unmet summed over the four-week planning horizon, not a continuous bed-day integral. Constraints:

$$z_{rr,h}^m + \sum_{r' \neq r: T_{rr'} \leq \tau} z_{rr',h}^m + u_{r,h}^m \geq \xi_{r,h}^m \quad \forall r, h, \quad (6)$$

$$z_{rr,h}^m + \sum_{r' \neq r: T_{r'r} \leq \tau} z_{r'r,h}^m \leq \Gamma_r + b_r \quad \forall r, h, \quad (7)$$

$$\sum_{r' \neq r} z_{rr',h}^m \leq \kappa \quad \forall r, h, \quad (8)$$

$$\sum_r b_r \leq B, \quad (9)$$

$$z_{rr',h}^m = 0 \quad \text{if } T_{rr'} > \tau, \quad (10)$$

$$b_r, z_{rr',h}^m, u_{r,h}^m \geq 0. \quad (11)$$

Equation (6) is a demand balance in which each region's forecast demand is served locally ($z_{rr,h}^m$), transferred to a reachable region ($z_{rr',h}^m$, $r' \neq r$), or left unmet ($u_{r,h}^m$). Equation (7) caps the beds a region uses (its own local patients plus those transferred in) at baseline plus surge. Equation (8) bounds the patients a region can sustain away from home by its transfer-service capacity κ ; Equation (9) charges the budget once per region (the same physical beds serve every horizon); Equation (10) suppresses transfers beyond the mutual-aid window. Because $u_{r,h}^m$ is minimised, Equation (6) holds it at the positive part of the residual shortfall, $\max(0, \xi_{r,h}^m - \sum_{r'} z_{rr',h}^m)$, so unmet demand is a derived quantity rather than a free decision.

3.3 Tail-Risk Robust Objective

To hedge against the high-demand scenario the robust model imposes Equations (6) to (8) for every $s \in \mathcal{S}$, retains Equations (9) and (10), and adds the tail-risk term $\lambda_3 W$ where $W \geq \sum_{r,h} u_{r,h}^s$ for $s \in \mathcal{S}_{\text{tail}} = \{\text{high}\}$:

$$\min \sum_{s \in \mathcal{S}} \pi_s \left(\sum_{r,h} u_{r,h}^s + \theta \sum_{r \neq r',h} \ell_{rr'} z_{rr',h}^s \right) + \lambda_3 W. \quad (12)$$

$\lambda_3=0$ recovers the three-scenario risk-neutral objective; $\lambda_3 \rightarrow \infty$ yields a worst-case policy on the high scenario. We report $\lambda_3=1$ throughout. With a single tail scenario, minimisation drives $W = \sum_{r,h} u_{r,h}^{\text{high}}$, so $\lambda_3 W$ is a linear penalty on the worst-scenario shortfall (equivalently, extra weight on the high scenario) rather than a Rockafellar–Uryasev CVaR; the auxiliary form generalises unchanged to multiple tail scenarios. The three-scenario construction is also co-monotone across regions and horizons (the high scenario imposes $q^{(0.9)}$ jointly), so the resulting tail is over-conservative relative to the true joint distribution; we treat this as an explicit stress-test design choice and examine its efficiency cost in Section 5.3.

3.4 Exact LP Solution and Baselines

Both the deterministic and robust models are linear programs (7 surge variables, 84 local-service self-loops, up to 504 inter-regional transfer flows, 84 unmet

variables, and the tail auxiliary, all continuous). A deployed plan rounds the resulting per-region surge allocations to integers, which changes total coverage by at most a few beds at this budget. We solve them exactly in PuLP with bundled CBC; at seven-region scale each instance reaches global optimality in <0.1 s, so metaheuristic search is deferred to the trust-level extension (Section 6).

Every policy is scored on equal footing by a common LP re-evaluation. For a fixed surge $\mathbf{b}^* \in \mathbb{R}_+^7$ with $\sum_r b_r^* \leq B$ and $b_r^* \leq K_r$, the residual problem in (z, u) is an LP that fills in transfers and unmet demand across all three scenarios, giving honest $E[u]$ and u^{worst} for the deterministic, robust, and heuristic allocations alike.

Finally, three closed-form heuristics give a no-optimisation floor: population-proportional uses ONS region populations, demand-proportional uses expected peak demand, and greedy shortage-first adds single-bed increments to the currently most-unmet region. Each fixes $\mathbf{b}^*$, which the same LP step then completes.

4 NHS England Case Study

The case study uses two public sources covering 1 Aug 2020 – 31 Aug 2022 (the period of daily NHS regional publication): NHS England COVID-19 Hospital Activity [12], which provides daily admissions, total hospital cases, total occupied beds, and mechanical-ventilation bed occupancy at NHS regional resolution; and ONS Mid-Year Population Estimates [13] on 2021 geography, used both as the per-region PINN normaliser and as the denominator of the population-proportional allocation baseline. All experiments run on the seven NHS England commissioning regions. The series is split chronologically by epidemic wave so the most clinically distinct wave is held out: training on 1 Aug 2020 – 31 May 2021 (Alpha and early vaccination), validation on 1 Jun – 30 Nov 2021 (Delta) for hyper-parameter selection and early stopping, and testing on 1 Dec 2021 – 31 Aug 2022 (Omicron and post-Omicron). Omicron’s case-to-hospitalisation and hospitalisation-to-critical-care ratios differ markedly from earlier waves, providing a regime-shift stress test.

For the allocation problem, baseline capacity Γ_r is *measured* as each region’s Delta-wave peak MV-bed occupancy $\times 1.05$, whereas the per-region surge cap $K_r = 0.5 \Gamma_r$, total budget $B = 0.20 \sum_r \Gamma_r \approx 213$ beds (the surge headroom above the $\sum_r \Gamma_r \approx 1065$ -bed measured baseline, not existing beds), mutual-aid travel cap $\tau = 240$ min, and uncapped outbound transfers ($\kappa = \infty$) are planning assumptions. The surge b_r is the model’s decision, the extra MV beds a region stands up above its measured baseline. The forecaster, budget B , tail weight λ_3 , travel cap τ , and transfer cap κ are each swept in Section 5.3. Forecast metrics are macro-averaged across the seven regions and the 32 rolling-origin evaluations spanning the Omicron test period; the full solver comparison is reported at the first Omicron decision origin (29 Dec 2021), with an all-origin check in Section 5.3. Code, configurations, and the cleaned dataset are available in the project repository.

Table 1. Forecast accuracy (best **bold**, second best underlined per column).

Model	RMSE by horizon (days)				MASE
	7	14	21	28	(all h)
Seasonal-naive(7)	13.53	18.16	<u>22.31</u>	25.98	0.63
ARIMA per region	11.33	17.80	23.41	<u>28.02</u>	<u>0.61</u>
XGBoost (lag features)	12.45	22.90	36.42	58.88	0.89
GRU per region	14.13	22.92	33.35	43.97	0.90
PinnGRU (raw)	<u>11.02</u>	<u>14.46</u>	24.19	34.63	0.77
PinnGRU+recal (ours)	10.56	13.34	20.79	28.54	0.56

5 Results and Discussion

5.1 Forecasting Accuracy

Table 1 reports per-horizon RMSE and MASE against four baselines under a common 32-origin rolling-origin protocol; MASE scales the mean absolute error by the in-sample seasonal-naive(7) MAE, so values below 1 beat that benchmark. The *raw* PinnGRU point forecast (frozen PINN encoder of Section 2.2 with the GRU head of Section 2.3) wins $h=7$ and $h=14$; a paired Wilcoxon signed-rank test across the 32 origins confirms the $h=14$ gain over ARIMA (-3.88 MV beds, $p=0.003$). It degrades at long horizons, however, where a regime-shift over-prediction bias (an $h=28$ underestimation rate of only 10.3%) leaves it behind ARIMA and seasonal-naive ($+5.53$ MV beds vs ARIMA at $h=28$, $p=0.006$). The leakage-free per-origin recalibration of Section 2.5 removes this bias and blends with persistence at long horizons. The recalibrated model is best at $h=7$, 14 and 21, cutting the operational 14-day RMSE by 25% over ARIMA and 42% over the per-region GRU, and at $h=28$ ties ARIMA (28.5 vs 28.0) with only seasonal-naive (26.0) still ahead. Its $h=28$ underestimation rate rises to 37%, so the point forecast is now a well-calibrated median rather than a systematic over-predictor.

XGBoost and the per-region GRU, lag-only learners bounded by the 28-day window, cannot extrapolate into Omicron’s altered case-to-hospitalisation regime and degrade sharply ($h=28$ RMSE 58.9, 44.0); ARIMA’s per-origin refit degrades modestly, and the recalibrated PinnGRU matches it across horizons.

Three ablations against the raw PinnGRU ($h=14$ RMSE 14.46, MASE 0.77) isolate the load-bearing parts: replacing the pinball loss with MSE lifts $h=14$ RMSE to 18.70 (MASE 0.97), skipping PINN pre-training to 17.27 (MASE 0.76), and zeroing the four PINN parameter features to 27.25 (MASE 1.23), the largest regression. The physics prior, not capacity, drives the gain over the per-region GRU.

5.2 Allocation Performance Across Solvers

Table 2 compares the allocation policies on the seven-region problem with $B = 213$ beds (20% of total baseline), four-hour mutual-aid cap, and PinnGRU

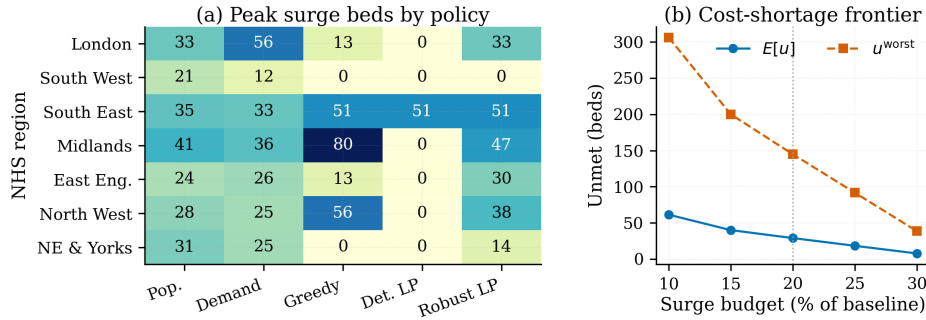
Table 2. Allocation comparison under $B = 213$ beds, $\tau = 240$ min, and three PinnGRU scenarios; the $\kappa=10$ column re-scores each policy under a capped transfer service.

Policy	Beds	$\kappa=\infty$		$\kappa=10$	Transfer (patient km)	Time (s)
		$E[u]$	u^{worst}	$E[u]$		
Population-proportional	212.7	29.0	144.8	48.8	3119.8	0.00
Demand-proportional	212.7	29.0	144.8	51.4	2615.3	0.00
Greedy shortage-first	212.7	29.0	144.8	32.3	1140.0	9.98
Deterministic LP	51.4	83.3	416.4	84.4	1175.1	0.03
Robust LP ($\lambda_3=1$)	212.7	29.0	144.8	32.3	671.7	0.04

$q^{(0.1)}/q^{(0.5)}/q^{(0.9)}$ forecasts at the first Omicron-period origin; every allocation is re-scored by the common LP re-evaluation over the full three-scenario set, so $E[u]$ and u^{worst} are on equal footing.

The *robust LP* achieves the lowest transfer burden (672 patient-km, $4.6\times$ below population-proportional and $1.7\times$ below greedy), but at $\kappa=\infty$ this is a tie-break among policies with identical rounded $E[u]$ and u^{worst} , not a coverage gain. The *deterministic LP* spends only 51 of the 213 beds and pays $\sim 3\times$ higher $E[u]$ and u^{worst} , quantifying the value of π_s -weighted expectation over median-only optimisation. The $\kappa=10$ column previews Section 5.3, where a capped transfer service breaks the tie.

The per-region allocations underlying Table 2 (Figure 3a) show the deterministic LP surging only Midlands and South East (where median demand exceeds baseline), whereas budget-exhausting policies cover those two and differ on the rest, the robust LP compromising with non-trivial North West and London surge.

**Fig. 3.** Allocation structure and robust-LP budget frontier: (a) per-region surge beds by policy at the first Omicron origin; (b) expected and worst-case unmet demand as the surge budget grows.

5.3 Forecast-Quality and Parameter Sensitivity

Substituting each forecaster into the robust LP at the 29 Dec 2021 origin, the $q^{(0.9)}$ peak sums (PinnGRU 1421, XGBoost 1105, GRU 973, ARIMA 875, seasonal-naive 745 beds) all exceed the realised peak (797) except seasonal-naive. Realised unmet is zero throughout, so forecast quality buys *efficiency* at the $q^{(0.9)}$ margin rather than coverage in this mild Omicron instance.

Across all 32 Omicron origins the pattern holds. Robust LP, population-proportional, and demand-proportional allocations share a mean expected unmet of 2.9, but the robust LP cuts mean transfer burden to 76 patient-km versus 406 and 337. The proportional baselines fill the full 212.7-bed budget at every origin, whereas the robust LP scales surge to local demand (mean 35.4 beds), so the routing-efficiency gain holds across the full demand distribution.

Turning to the planning parameters, a one-at-a-time sweep isolates each control. The budget B dominates (Figure 3b). Cutting $B/\sum_r \Gamma_r$ from 0.20 to 0.10 doubles $E[u]$ and triples u^{worst} , while 0.30 cuts $E[u]$ to $\sim 1/4$. λ_3 is *non-binding* at $B = 0.20$ (the expectation-minimising solution already clears the high tail); only a tighter budget ($\leq 15\%$) re-engages it.

The transfer controls τ and κ determine when placement itself matters. With uncapped recourse on the 12-link network the Table 2 coverage tie persists at budgets of 10–20% because coverage depends only on *total* surge; placement is fungible and optimisation buys routing alone. Constraining recourse breaks the tie. Pruning the network to five links ($\tau=120$ min) puts the robust LP ahead of every heuristic on expected and worst-case unmet (29.9 vs 32.3–34.9 at $B=20\%$), and capping the transfer service at $\kappa=10$ produces the same separation (the $\kappa=10$ column of Table 2), where greedy matches the LP only by issuing 213 calls to the same LP oracle at $1.4\times$ its transfer burden. Placement earns its keep when transfers are scarce, the operating regime of a real surge.

6 Conclusion

We presented a forecast-to-decision pipeline for NHS critical-care surge planning that couples a per-region physics-informed forecaster to a tail-risk robust allocation linear program. With a leakage-free per-origin recalibration, PinnGRU is competitive across the horizons that drive surge decisions, cutting 14-day RMSE by 25% over refitted ARIMA and 42% over a neural GRU. On the decision side the gain is structural. With unconstrained transfers every budget-exhausting policy ties on coverage and the robust formulation wins on routing; once transfers are scarce the tie breaks and the robust LP is never beaten on coverage. The main caveats are detour-scaled travel times, well-mixed populations, scaled cost units, and a single realised origin; more deeply, the per-region forecaster and co-monotone scenarios over-state the joint tail and understate transfers, which pay off most under *asynchronous* regional peaks. Correlated demand scenarios and a trust-level instance where exact solution warrants metaheuristics are the natural next steps.

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